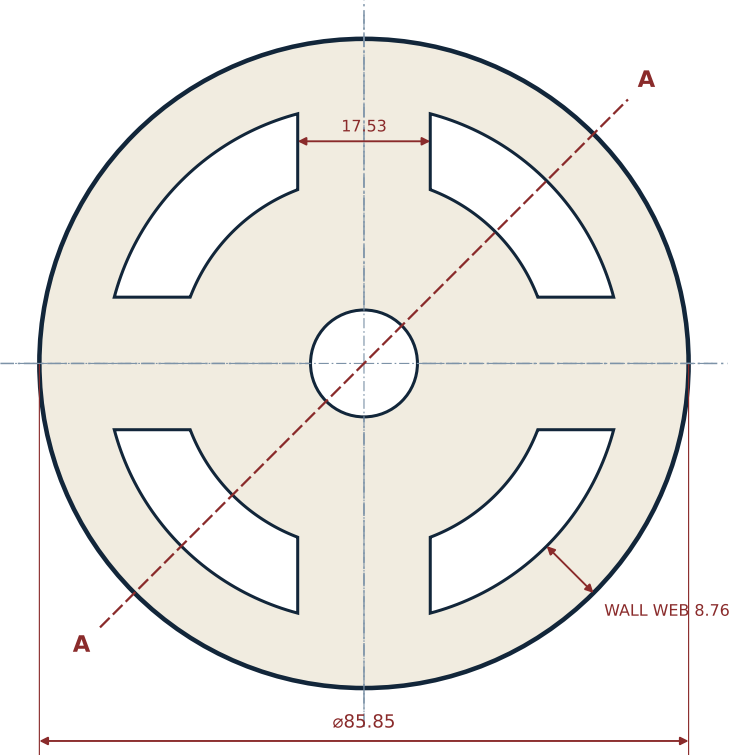


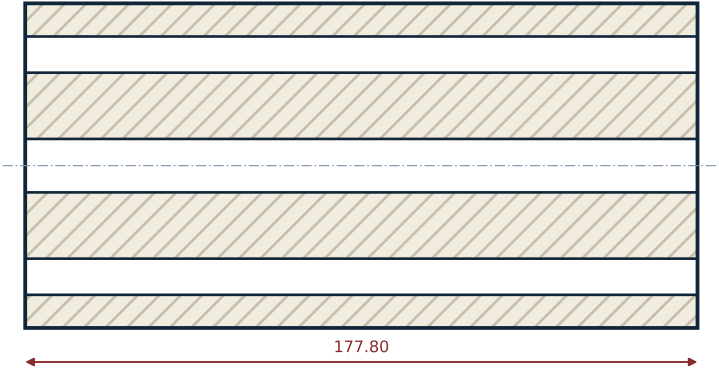
PORTS

CENTRE PORT $\phi 14.14$
4 SECTOR PORTS R24.60 TO R34.16, EQ SP (90° PITCH)
SPOKES 17.53 THK, RING WEB 17.53, WALL WEB 8.76



CROSS-SECTION FROM FORWARD END SCALE 1 : 1

SECTION A-A SCALE 1 : 2



DESIGN DATA

PORT AREA	1241.4 mm ²
BURNING PERIMETER	351.3 mm
BURNING SURFACE	625 cm ²
WALL WEB (TO CASE)	8.76 mm
WEB BETWEEN PORTS	17.53 mm
PORT / THROAT AREA	1.70
FUEL MASS	0.728 kg

NOTES

- ALL DIMENSIONS IN MILLIMETRES. ORIGIN ON THE GRAIN AXIS. CROSS-SECTION IS VIEWED FROM THE FORWARD (INJECTOR) END. PORTS RUN THE FULL LENGTH.
- SPOKES ARE CONSTANT THICKNESS: EACH SECTOR SIDE IS PARALLEL TO ITS SPOKE CENTRELINE, NOT RADIAL. INNER AND OUTER PORT WALLS ARE ARCS ON THE GRAIN AXIS.
- PORT CORNERS ARE DRAWN SHARP. CAST WITH R2 FILLETS TO AVOID CRACKING; THIS REMOVES ABOUT 14 mm² (1.1%) OF PORT AREA.
- PRINT AT 100% ON A4 LANDSCAPE FOR TRUE SCALE — CONFIRM WITH THE CHECK BAR BEFORE MEASURING OFF THIS SHEET.
- RUN VALVE CLOSES AT 3.8 s. WITH THE LIKELY-N2O REGRESSION FITS (SP7) THE PORTS REACH 6.2-7.1 mm OF THE 8.76 mm WALL WEB. REGRESSION COEFFICIENTS ARE NOT MEASURED ON THIS MOTOR.

CHECK BAR — 50 mm ON PAPER:

FUEL GRAIN

PROJECT	V2.2 N2O / PARAFFIN
PART NO	—
MATERIAL	PARAFFIN
GRAIN	$\phi 85.85 \times 177.80$
PORTS	CENTRE + 4 SECTOR
SCALE	1 : 1 / 1 : 2 UNITS mm
DRAWN	—
HASTE	2026-09-29